Systems of Linear Inequalities

Algebra Worksheet · Grade 9–11

Name: ________________________________ Date: __________________

Learning Objectives

- Graph a single linear inequality using solid or dashed boundary lines and correct shading
- Solve a linear inequality for y and identify the appropriate shading region
- Find the solution region of a system of two or more linear inequalities by graphing

Problems

1. Which type of boundary line (solid or dashed) should be used when graphing each inequality? State your answer for each.

$$\begin{cases} y > 3x + 1 \\ y \le -2x + 4 \\ y \ge x - 5 \\ y < \frac{1}{2}x + 2 \end{cases}$$

2. For each inequality below, state whether you shade above or below the boundary line.

$$\begin{cases} y < 4x - 2 \\ y \ge -x + 6 \\ y > \frac{2}{3}x \\ y \le 5 - 3x \end{cases}$$

3. Graph the inequality on a coordinate plane. Identify the y-intercept, slope, type of line, and shading direction.

$$y < 2x - 3$$

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4. Graph the inequality on a coordinate plane. Identify the y-intercept, slope, type of line, and shading direction.

$$y \geq -\frac{1}{3}x + 4$$

5. Solve the inequality for y, then state the slope, y-intercept, line type, and shading direction.

$$3x + y > 9$$

6. Solve the inequality for y, then identify the slope, y-intercept, line type, and shading direction.

$$2x + 3y \leq 12$$

7. Solve the inequality for y, then describe how to graph it including line type and shading.

$$4x - 2y > 8$$

8. Graph the system of linear inequalities and identify the solution region (shade and describe where the overlapping region lies).

$$\begin{cases} y < 2x - 3 \\ y > x - 1 \end{cases}$$

9. Solve each inequality for y, then graph the system and describe the solution region.

$$\begin{cases} \mathbf{2x + y \leq 6} \\ x - y < 1 \end{cases}$$

10. Graph the system of three linear inequalities and describe the shape and location of the solution region.

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$$\begin{cases} y \leq -x + 5 \\ y \geq \frac{1}{2}x - 1 \\ x \geq 0 \end{cases}$$

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Systems of Linear Inequalities — Answer Key

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Answer Key

1. Answer: Dashed; Solid; Solid; Dashed

- Strict inequalities ($>$ or $<$) use a dashed line because the boundary is NOT included.
- Non-strict inequalities ($\geq$ or $\leq$) use a solid line because the boundary IS included.
- $y > 3x+1$: dashed | $y \leq -2x+4$: solid | $y \geq x-5$: solid | $y < (1/2)x+2$: dashed

2. Answer: Below; Above; Above; Below

- If inequality is $<$ or $\leq$, shade BELOW the line.
- If inequality is $>$ or $\geq$, shade ABOVE the line.
- $y < 4x-2$: below | $y \geq -x+6$: above | $y > (2/3)x$: above | $y \leq 5-3x$: below

3. Answer: y-intercept: -3; slope: 2; dashed line; shade below

- Identify slope $m = 2$ and y-intercept $b = -3$.
- Plot the y-intercept at $(0, -3)$.
- From $(0, -3)$ move 2 units up and 1 unit right to plot a second point at $(1, -1)$.
- Draw a dashed line through both points (strict inequality $<$).
- Shade the region below the line (less than symbol).

4. Answer: y-intercept: 4; slope: -1/3; solid line; shade above

- Identify slope $m = -1/3$ and y-intercept $b = 4$.
- Plot the y-intercept at $(0, 4)$.
- From $(0, 4)$ move 1 unit DOWN and 3 units to the right to plot $(3, 3)$.
- Draw a solid line ($\geq$ includes the boundary).
- Shade the region above the line (greater than or equal to).

5. Answer: $y > -3x + 9$; slope: -3; y-intercept: 9; dashed line; shade above

- Subtract $3x$ from both sides: $y > -3x + 9$.
- Slope $m = -3$, y-intercept $b = 9$.
- Use a dashed line because the symbol is strictly $>$.
- Shade above the line because the symbol is $>$.

6. Answer: $y \leq -2/3\ x + 4$; slope: -2/3; y-intercept: 4; solid line; shade below

- Subtract $2x$ from both sides: $3y \leq -2x + 12$.
- Divide every term by 3: $y \leq -(2/3)x + 4$.
- Slope $m = -2/3$, y-intercept $b = 4$.
- Use a solid line ($\leq$ includes the boundary).
- Shade below the line (less than or equal to).

7. Answer: $y < 2x - 4$; slope: 2; y-intercept: -4; dashed line; shade below

- Subtract $4x$ from both sides: $-2y > -4x + 8$.

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- Divide every term by -2 (flip the inequality sign!): $y < 2x - 4$.
- Slope $m = 2$, y-intercept $b = -4$.
- Use a dashed line (strict inequality <).
- Shade below the line.

8. Answer: The solution is the overlapping region below the dashed line $y = 2x - 3$ and above the dashed line $y = x - 1$.

- Graph $y = 2x - 3$ as a dashed line (y-intercept -3, slope 2). Shade below.
- Graph $y = x - 1$ as a dashed line (y-intercept -1, slope 1). Shade above.
- The solution region is where both shaded areas overlap — below the first line and above the second.
- The two boundary lines intersect at (2, 1); the solution region is to the right of this point.

9. Answer: $y \leq -2x + 6$ (solid, shade below); $y > x - 1$ (dashed, shade above); solution is overlapping region above dashed line and below solid line.

- Solve first inequality: $y \leq -2x + 6$. Solid line, shade below.
- Solve second inequality: subtract x $\rightarrow$ $-y < -x + 1$ $\rightarrow$ $y > x - 1$. Dashed line, shade above.
- Graph both lines on the same coordinate plane.
- The solution region is where both shaded areas overlap.

10. Answer: Triangular solution region in the first quadrant bounded by $x = 0$ on the left, below the line $y = -x + 5$, and above the line $y = (1/2)x - 1$.

- Graph $y = -x + 5$ as a solid line (y-intercept 5, slope -1). Shade below.
- Graph $y = (1/2)x - 1$ as a solid line (y-intercept -1, slope 1/2). Shade above.
- Graph $x = 0$ (the y-axis) as a solid vertical line. Shade to the right ($x \geq 0$).
- Find vertices: Intersection of $y = -x+5$ and $y = (1/2)x-1$ $\rightarrow$ $x = 4$, $y = 1$, point (4,1). Intersection of $y = -x+5$ and $x = 0$ $\rightarrow$ (0,5). Intersection of $y = (1/2)x-1$ and $x = 0$ $\rightarrow$ (0,-1).
- The solution region is the triangle with vertices (0, -1), (0, 5), and (4, 1) — the area satisfying all three inequalities simultaneously.

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